



Problem-Solution Fit

Date	8 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Bank credit officers and financial analysts who evaluate lending risks daily, alongside applicants seeking quick loan eligibility decisions.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Branch personnel lack technical data science expertise. The system must run securely on existing desktop terminals and field tablets without needing specialized hardware upgrades.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual underwriting allows thorough physical verification but is slow and unscalable. Legacy scoring sheets enforce rules consistently but fail to capture complex, non-linear data patterns.
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2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]

Daily manual underwriting bottlenecks cause long processing delays, while a lack of automated screening increases human error and the risk of non-performing assets (NPAs).

9. PROBLEM ROOT / CAUSE [RC]

Lending datasets frequently contain missing continuous and categorical values, which traditional legacy infrastructures cannot preprocess or evaluate using advanced machine learning models.

7. BEHAVIOR

+ ITS INTENSITY [BE]

Analysts experience high stress during peak cycles as they cross-reference disparate spreadsheets and manually track missing information.

3. TRIGGERS TO ACT [TR]

High-volume application periods that overwhelm manual workflows, coupled with strict institutional mandates to reduce loan default rates.

10. YOUR SOLUTION [SL]

Smart Lender: A web-based application built with Python and Flask, hosted securely on IBM Cloud. It automates data imputation and utilizes a pre-trained XG Boost model (94.7% training / 81.1% testing accuracy) to deliver instant approval or high-risk predictions.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE Legacy internal core-banking terminals and basic data entry forms.

4. EMOTIONS

BEFORE / AFTER [EM]

Before: Staff feel overwhelmed by paperwork backlogs; applicants experience anxiety during long waiting periods
After: Officers feel confident using automated data insights; applicants experience satisfaction from instant decisions.

OFFLINE In-branch customer interviews and physical document sorting.